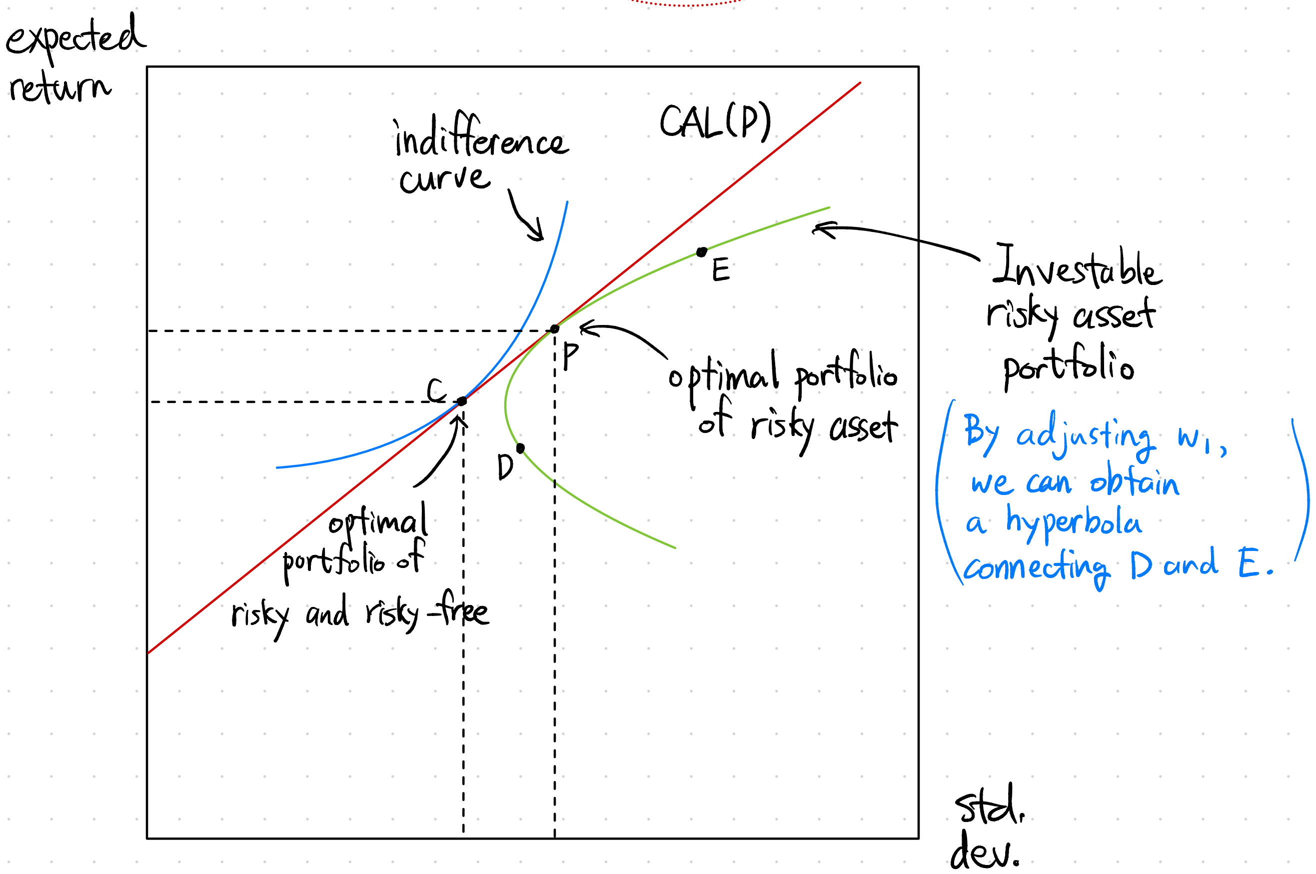
Rational Asset Allocation

Investors divide their assets into two broad categories for management.

Assets
- Risky-free Assets (e.g. bank deposit)
- Risky Assets (e.g. Stocks, Bonds)
  - Stock (E)
  - Bond (D)

Capital Allocation

Asset Allocation between stocks and bonds.



The part of hyperbola between D and E is a curve obtained where $0 \le w_1, w_2 \le 1$ (100%)

If w_1 or w_2 exceed 100%, then we obtain a curve outside of $\overline{DE}$

That is, another one is less than 0 and it implies "short selling."

$$\max_{W_1, W_2} \frac{\{E(r_P) - r_f\}}{\sigma_P}$$

$$\Rightarrow W_1 = \frac{\{E(r_A) - r_f\}\sigma_B^2 - \{E(r_A) - r_f\}\,Cov(r_A, r_B)}{\{E(r_A) - r_f\}\sigma_B^2 + \{E(r_B) - r_f\}\sigma_A^2 - \{E(r_A) + E(r_B) - 2r_f\}\,Cov(r_A, r_B)}$$

$$W_2 = 1 - W_1$$

	Bond	Stock
Expected Return	5 %	9 %
Standard Deviation	13 %	23 %
Covariance	63.25	
Correlation	0.25	

Optimal Risky Asset Portfolio

$W_A = 43\%$, $W_B = 57\%$

$\rightsquigarrow E(r_p) = 0.43 \times 5 + 0.57 \times 9 = 7.28\ \%$

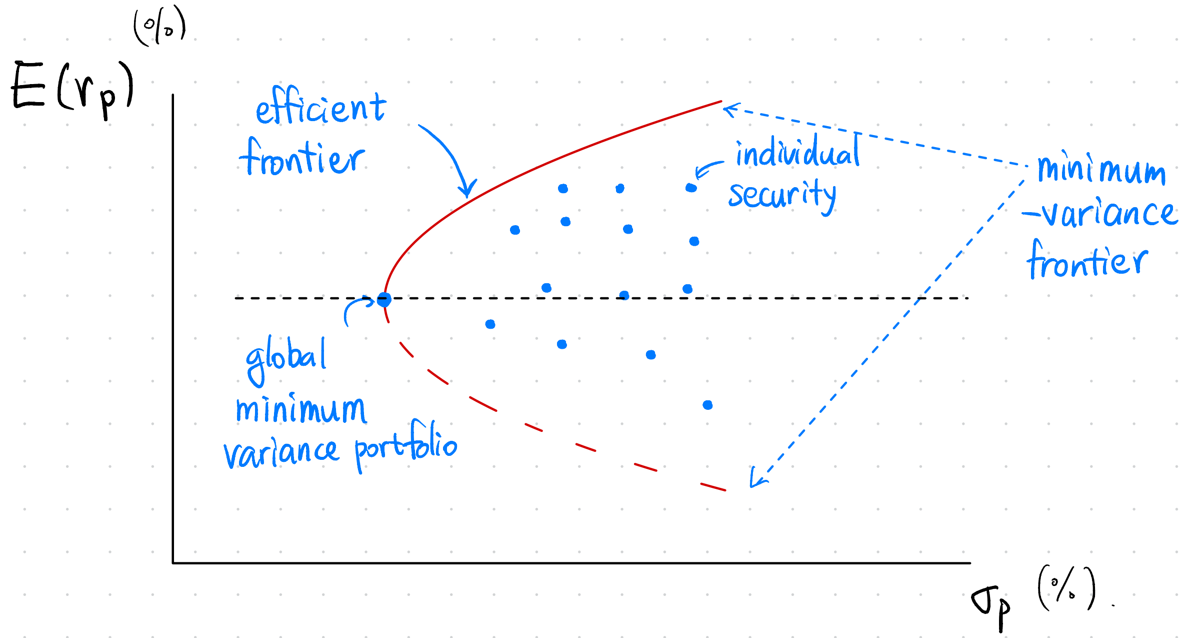
$\sigma(r_p) = \sqrt{0.43^2 \times 13^2 + 0.57^2 \times 23^2 + 2 \times 0.43 \times 0.57 \times 63.25} = 15.30\ \%$

Let risk aversion coefficient $A = 3$ and $r_f = 3\%$.

$$y = \frac{E(r_p) - r_f}{A\sigma_p^2} = 60.95\%$$

$\Rightarrow$ asset D weight : $0.43 \times 0.6095 = 26.21\%$

asset E weight : $0.57 \times 0.6095 = 34.74\%$



Asset Allocation and Diversification

- According to Markowitz's theory, a rational investor invest in the same portfolio of risky assets regardless of their risk aversion coefficient.

- Construction of the optimal portfolio is often considered to consist of two independent problems. This concept, fundamentally derived from the Two-Fund Separation Theorem (or separation principle)

1. Allocating between the risky portfolio and the risk-free asset (Capital Allocation)
2. Determining the optimal risky portfolio (Asset Allocation)

Diversification

- generally, the portfolio diversification is:

$$\sigma_p^2 = \sum_{i=1}^{n}\sum_{j=1}^{m} W_i W_j \, Cov(r_i, r_j)$$

- Equally weighted portfolio $(W_i = \frac{1}{n})$

$$\sigma_p^2 = \sum_{i=1}^{n}\sum_{j=1}^{n}\left(\frac{1}{n}\right)\left(\frac{1}{n}\right) Cov(r_i, r_j)$$

$$= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} Cov(r_i, r_j)$$

Divide two cases. $\begin{cases} ① \; i=j \\ ② \; i\neq j \end{cases}$

$$= \frac{1}{n^2}\left[\sum_{i=1}^{n}\sigma_i^2 + \sum_{i=1}^{n}\sum_{j\neq i}^{n} Cov(r_i, r_j)\right]$$

① $i=j \Rightarrow Cov(r_i, r_i) = \sigma_i^2$

② $i\neq j \Rightarrow n^2 - n = n(n-1)$ terms.

$$= \frac{1}{n^2}\left[n\overline{\sigma}^2 + n(n-1)\overline{Cov}\right]$$

where $\begin{cases} \overline{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\sigma_i^2 \\ \overline{Cov} = \frac{1}{n(n-1)}\sum_{i=1}^{n}\sum_{i\neq j} Cov(r_i, r_j) \end{cases}$

$$\therefore \; \sigma_p^2 = \frac{1}{n}\overline{\sigma}^2 + \frac{n-1}{n}\overline{Cov}$$

$\frac{1}{n}\overline{\sigma}^2$ (Idiosyncratic Risk, Specific Risk, Diversifiable Risk)

: converges to 0, when $n\uparrow$. That is, it is removable when diversify~.

$\frac{n-1}{n}\overline{Cov}$ (Systematic Risk, Market Risk, Undiversifiable Risk)

: $\frac{n-1}{n} \rightarrow 1$ when $n\uparrow$, so $\overline{Cov}$ only remains.

$\Rightarrow$ " Systematic Risk
Undiversifiable Risk "